Chapter 2

Biological mechanisms of reactive oxygen species (ROS)

Reactive oxygen species (ROS) is a generic name given to a variety of molecules and free radicals derived from molecular oxygen. The reduction of oxygen produces relatively stable intermediates. One-electron reduction produces a superoxide anion, which is the precursor of most ROS. As most commonly used, ROS in this chapter refer to superoxide, hydrogen peroxide, and their derivatives such as the hydroxyl radical.

ROS were initially recognized as toxic by-products of aerobic metabolism. In recent years, it has become apparent that ROS play an important signaling role in plants, controlling processes such as growth, development, and especially response to biotic and abiotic environmental stimuli.

ROS include free radicals such as superoxide anion $\left(O_2^{\bullet-}\right)$, hydroxyl radical ($^{\bullet}OH$), as well as nonradical molecules like hydrogen peroxide (H_2O_2), singlet oxygen ($^1O_2$), and so forth. Stepwise reduction of molecular oxygen (O_2) by high-energy exposure or electron-transfer reactions leads to the production of ROS that are highly reactive.

ROS are produced by living organisms as a result of normal cellular metabolism and environmental factors, such as air pollutants or cigarette smoke. ROS are highly reactive molecules and can damage cell structures such as carbohydrates, nucleic acids, lipids, and proteins and alter their functions. At low to moderate concentrations, they function in physiological cell processes, but at high concentrations, they produce adverse modifications to cell components, such as lipids, proteins, and DNA [1–6].

ROS are key signaling molecules that play an important role in the progression of inflammatory disorders. Enhanced ROS generation by polymorphonuclear neutrophils (PMNs) at the site of inflammation causes endothelial dysfunction and tissue injury. The vascular endothelium plays an important role in the passage of macromolecules and inflammatory cells from the blood to tissue. Under inflammatory conditions, oxidative stress produced by PMNs leads to the opening of inter-endothelial junctions and promotes the migration of inflammatory cells across the endothelial barrier. The migrated inflammatory cells not only help in the clearance of pathogens and foreign particles but also lead to tissue injury.

ROS (e.g., $^1O_2, O_2^{\bullet-}, H_2O_2, ^{\bullet}OH, OH^-$) are partially reduced or activated forms of atmospheric oxygen (O_2). They are considered to be unavoidable by-products of aerobic metabolism that have accompanied life on Earth ever since the appearance of oxygen-evolving photosynthetic organisms about 2.2–2.7 billion years ago [7]. Higher plants have thus evolved in the presence of ROS and have acquired dedicated pathways to protect themselves from ROS toxicity, as well as to use ROS as signaling molecules [8–12]. If kept unchecked, ROS concentrations will increase in cells and cause oxidative damage to membranes (lipid peroxidation), proteins, RNA and DNA molecules, and even lead to the oxidative destruction of cells in a process termed oxidative stress [13]. However, this process is mitigated in cells by a large number of ROS detoxifying proteins (e.g., superoxide dismutase [SOD], ascorbate peroxidase, catalase, glutathione peroxidase [GPx], and peroxiredoxin [Prx]), as well as by antioxidants such as ascorbic acid and glutathione (γ-glutamyl-cysteinyl-glycine; GSH) that are present in almost all subcellular compartments [14]. The active process of ROS detoxification in plant cells is also aided by different metabolic adaptations that reduce ROS production, and by maintaining the level of free transient metals such as Fe^{2+} under control, to prevent the formation of the highly toxic hydroxyl radical ($^{\bullet}OH$) via the Fenton reaction [15]. On the other hand, plants actively produce ROS that are used as signal transduction molecules.

ROS $\left(^1O_2, O_2^{\bullet-}, H_2O_2, ^{\bullet}OH, OH^-\right)$ are highly reactive, toxic molecules that are caused by various environmental stress conditions, increased oxidative stress, and which cause damage to membranes, DNA, protein, photosynthetic pigments, and lipids [16, 17].

ROS are deadly weapons used by phagocytes and other cell types, such as lung epithelial cells, against pathogens. ROS can kill pathogens directly by causing oxidative damage to biocompounds or indirectly by stimulating pathogen elimination by various nonoxidative mechanisms, including pattern recognition receptor signaling, autophagy, neutrophil extracellular

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trap formation, and T-lymphocyte responses. Thus one should expect that the inhibition of ROS production promotes infection. Increasing evidence supports that in particular infections, antioxidants decrease and prooxidants increase pathogen burden. In this study, we review the classic infections that are controlled by ROS and the cases in which ROS appear as promoters of infection, challenging the paradigm. We discuss the possible mechanisms by which ROS could promote particular infections. These mechanisms are still not completely clear but include the metabolic effects of ROS on pathogen physiology, ROS-induced damage to the immune system, and ROS-induced activation of immune defense mechanisms that are subsequently hijacked by particular pathogens to act against more effective microbicidal mechanisms of the immune system.

ROS are used by the immune system as weapons against pathogens; however, antioxidants have long been recognized as protectors of host organisms against infections [18]. It is plausible that the paradox contained in these statements can be solved by the failure of antioxidants to neutralize the ROS involved in pathogen killing (e.g., by not reaching the appropriate location required for their action) and by the capacity of antioxidants to protect immune cells from the damage caused by ROS. The nature of the microbes and their susceptibility to ROS can also offer a clue to solving the paradox. ROS effectively combat certain microbes, whereas other microbes seem to thrive in oxidative environments. Recognizing the mechanisms by which ROS promote the clearance of microbes and the mechanisms by which particular microbes benefit from ROS generation will help to clarify the paradox.

Phagocytes reside within tissues or are recruited by inflammatory processes. Phagocytes recognize microbes through many molecular patterns displayed by them and that try to engulf them. Once a microbe is phagocytosed, the nature of the molecules recognized on the microbe's surface dictates the treatment enacted within the phagosome. Respiratory burst, a process by which nicotine adenine dinucleotide phosphate (NADPH) oxidase (NADPH oxidase-2 [NOX2]) generates ROS in response to microbe recognition, is a possible outcome of this process and helps to get rid of many microbes. For instance, β-glucan on the surface of the fungi can engage dectin-1 on the surface of the phagocyte [19]. Once fungi are phagocytosed, NOX2 is promptly assembled at the phagosome membrane and high amounts of superoxide $(O_2^{\bullet -})$ are discharged into the phagosome. The recognition of *Escherichia coli* generates comparatively smaller amounts of ROS than the recognition of *Listeria monocytogenes*, and fewer ROS are less promptly released into the phagosome [20]. However, the recognition and phagocytosis of *Leishmania* spp. is well studied, and, except when recognition is mediated by an Fc receptor, virulent parasites do not usually trigger a severe respiratory burst [21]. Thus microbes face different fates after phagocytosis dependent on the molecules presented at their surfaces and how they are targeted by innate immune receptors.

Once a pathogen is phagocytosed, it must subvert the respiratory burst, withstand its oxidative power, or escape the phagosome to survive. Microbe recognition sets the immune system in motion, and ROS are produced not only in the phagocyte respiratory burst but also in other cell compartments, such as mitochondria, and as intermediaries in many signal transduction pathways, such as leukocyte pattern recognition receptor signaling. The generation of ROS is a prerequisite to the formation of neutrophil extracellular traps [22] and is actively involved in phagolysosome formation and enzymatic degradation [23]; autophagy [24–26] and ROS inhibition of mammalian target of rapamycin (mTOR) kinase [27, 28]; chemoattraction and inflammation [29, 30]; cell death of infection reservoirs [31]; antigenic presentation, T-helper (Th) polarization, and lymphocyte proliferation [32–37]; iron redistribution among tissues [38] and cell compartment availability of iron [39–41]; and foam cell formation [42, 43]. Many of these mechanisms promote microbe clearance, whereas others can potentially contribute to microbe persistence.

2.1 Exogenous source of oxidants

2.1.1 Cigarette smoke

Cigarette smoke contains many oxidants, free radicals, and organic compounds, such as superoxide and nitric oxide (NO) [44]. In addition, inhalation of cigarette smoke into the lung also activates endogenous mechanisms, such as accumulation of neutrophils and macrophages, which further increase oxidant injury.

2.1.2 Ozone exposure

Ozone exposure can cause lipid peroxidation and induce influx of neutrophils into the airway epithelium. Short-term exposure to ozone also causes the release of inflammatory mediators, such as myeloperoxidase (MPO), eosinophil cationic proteins, and also lactate dehydrogenase and albumin [45]. Even in healthy subjects, ozone exposure causes a reduction in pulmonary functions [46]. Cho et al. [47] showed that particulate matter (a mixture of solid particles and liquid droplets suspended in the air) catalyzes the reduction of oxygen.

2.1.3 Hyperoxia

Hyperoxia refers to conditions of higher oxygen levels than normal partial pressure of oxygen in the lungs or other body tissues. It leads to greater production of reactive oxygen and nitrogen species [48, 49].

2.1.4 Ionizing radiation

Ionizing radiation, in the presence of O_2, converts hydroxyl radical, superoxide, and organic radicals to hydrogen peroxide and organic hydroperoxides. These hydroperoxide species react with redox active metal ions, such as Fe and Cu, via Fenton reactions and thus induce oxidative stress [50, 51]. Narayanan et al. [52] showed that fibroblasts that were exposed to alpha particles had significant increases in intracellular $O_2^{\bullet-}$ and H_2O_2 production via plasma membrane-bound NOX [52]. Signal transduction molecules, such as extracellular signal-regulated kinase-1 and -2, c-Jun N-terminal kinase, and p38, and transcription factors, such as activator protein-1, nuclear factor-κB (NF-κB), and p53, are activated, which result in the expression of radiation response-related genes [53–58]. Ultraviolet A (UVA) photons trigger oxidative reactions by excitation of endogenous photosensitizers, such as porphyrins, NOX, and riboflavins. 8-Oxo-7,8-dihydroguanine is the main UVA-mediated DNA oxidation product formed by the oxidation of $^{\bullet}OH$ radical, one-electron oxidants, and singlet oxygen that mainly reacts with guanine [59]. The formation of guanine radical cation in isolated DNA has been shown to efficiently occur through the direct effect of ionizing radiation [60, 61]. After exposure to ionizing radiation, intracellular level of GSH decreases for a short term but then increases [62].

2.1.5 Heavy metal ions

Heavy metal ions, such as iron, copper, cadmium, mercury, nickel, lead, and arsenic, can induce generation of reactive radicals and cause cellular damage via depletion of enzyme activities through lipid peroxidation and reaction with nuclear proteins and DNA [63].

One of the most important mechanisms of metal-mediated free radical generation is via a Fenton-type reaction. Superoxide ion and hydrogen peroxide can interact with transition metals, such as iron and copper, via the metal-catalyzed Haber-Weiss/Fenton reaction to form OH radicals:

$$\text{Metal}^{3+} + O_2^{\bullet-} \rightarrow \text{Metal}^{2+} + O_2 \quad \text{Haber-Weiss}$$
$$\text{Metal}^{2+} + H_2O_2 \rightarrow \text{Metal}^{3+} + OH^- + {}^{\bullet}OH \quad \text{Fenton reaction}$$

The role of oxidants in inducing inflammation has been vigorously investigated in all manner of experimental models. The consensus is that they are fundamentally involved, but how they contribute to the response and whether antioxidant therapy is a valid means of arresting inflammation in patients remain largely unresolved. Among the commonly used inflammatory mediators used to simulate inflammation are cytokines (e.g., tumor necrosis factor-α), the stress of hyperoxia, ischemia-reperfusion injury, bacterial toxins (lipopolysaccharide [LPS]), and mediators that ligate cell surface receptors (platelet activating factor, thrombin, histamine, vascular endothelial growth factor, and bradykinins). These and other mediators except LPS induce only a subset of changes that are associated with full-blown inflammation.

2.2 Endogenous sources of ROS and their regulation in inflammation

ROS are classically defined as partially reduced metabolites of oxygen that possess strong oxidizing capabilities. They are deleterious to cells at high concentrations but at low concentrations (exact concentrations still remain to be defined), they serve complex signaling functions. They are injurious, because they oxidize protein and lipid cellular constituents and damage the DNA. At "physiological concentrations," ROS function as signaling molecules that regulate cell growth, adhesion of cells toward other cells, differentiation, senescence, and apoptosis [64, 65]. The concept of chronic or prolonged ROS production is considered central to the progression of inflammatory disease [66]. What are biologically relevant ROS? The widely studied and understood family members are the superoxide anion ($O_2^{\bullet-}$), hydroxyl radical ($^{\bullet}OH$), hydrogen peroxide (H_2O_2), and hypochlorous acid (HOCl) [65]. Although others may be important in signaling and disease [65, 66], their functions remain poorly understood. ROS are generated as by-products of cellular metabolism through the electron transport chain (ETC) in mitochondria as well as via cytochrome P450. The other major source, where ROS are not produced as by-products, are the NOXs that are present in a variety of cells, especially the professional phagocytes and endothelial cells [67], which are central to the genesis of the inflammatory response [66].

$O_2^{\bullet-}$ is generated by one-electron reduction of O_2 through enzymatic catalysis by NOX or xanthine oxidase or during electron transfer reactions in the ETC of mitochondria [65, 68, 69]. $O_2^{\bullet-}$ has a half-life of 10^6 ns [70], as it undergoes spontaneous dismutation to H_2O_2 (under physiological conditions $k = 2 \times 10^5\ M^{-1}s^{-1}$). This reaction can be accelerated to 10^4-fold by the enzyme SOD ($k = 1.6 \times 10^9\ M^{-1}s^{-1}$) [65, 69]. In the presence of the transition metal iron, $O_2^{\bullet-}$ and H_2O_2, in turn, generate the highly reactive OH^- and $^{\bullet}OH$ (Haber-Weiss reaction). In the first step of this reaction, $O_2^{\bullet-}$ reacts with Fe^{3+} to form Fe^{2+} and O_2. However, this reaction is thermodynamically unfavorable under physiological conditions [71]. The second step of this reaction is also known as Fenton's reaction and occurs under the biological conditions in which Fe^{2+} reacts with H_2O_2 to form both $^{\bullet}OH$ and OH^-. $^{\bullet}OH$ is defined as the most potent oxidizing species of biological membrane proteins and lipids and has an extremely short half-life [65, 70, 72]. At inflammatory sites where PMN are abundant, H_2O_2 and chloride generate HOCl by the enzyme MPO, generally considered as being a PMN-specific enzyme [73]. The passage of $O_2^{\bullet-}$ across biological membranes is highly restricted because of its negative charge. However, certain transmembrane proteins, such as voltage-dependent anion channels (VDACs) found in mitochondria, allow transmembrane passage of $O_2^{\bullet-}$ produced in ETC [74]. H_2O_2, on the other hand, can cross biological membranes through aquaporin channels such as AQP3 and AQP8, which mediate membrane H_2O_2 uptake, raising the possibility that it can enter cells that are contacting one another [75–77].

ROS are produced from molecular oxygen as a result of normal cellular metabolism. ROS can be divided into two groups: free radicals and nonradicals. Molecules containing one or more unpaired electrons and thus giving reactivity to the molecule are called free radicals. When two free radicals share their unpaired electrons, nonradical forms are created. The three major ROS that are of physiological significance are superoxide anion ($O_2^{\bullet-}$), hydroxyl radical ($^{\bullet}OH$), and hydrogen peroxide (H_2O_2). ROS are summarized in Table 2.1.

Superoxide anion is formed by the addition of one electron to the molecular oxygen [78]. This process is mediated by NOX or xanthine oxidase or by the mitochondrial electron transport system. The major site for producing superoxide anion is the mitochondria, the machinery of the cell that produces adenosine triphosphate. Normally, electrons are transferred through the mitochondrial ETC for reduction of oxygen to water, but approximately 1%–3% of all electrons leak from the system and produce superoxide. NOX is found in polymorphonuclear leukocytes, monocytes, and macrophages. Upon phagocytosis, these cells produce a burst of superoxide that leads to bactericidal activity. Superoxide is converted into hydrogen peroxide by the action of SODs (EC 1.15.1.1). Hydrogen peroxide easily diffuses across the plasma membrane. Hydrogen peroxide is also produced by xanthine oxidase, amino acid oxidase, and NOX [79, 80] and in peroxisomes by consumption of molecular oxygen in metabolic reactions. In a succession of reactions called Haber-Weiss and Fenton reactions, H_2O_2 can breakdown to OH^- in the presence of transmission metals like Fe^{2+} or Cu^{2+} [81]:

$$O_2^{\bullet-} + Fe^{3+} \rightarrow O_2 + Fe^{2+} \quad \text{Haber-Weiss}$$
$$Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH^- + {}^{\bullet}OH \quad \text{Fenton reaction}$$

$O_2^{\bullet-}$ itself can also react with H_2O_2 and generate OH^- [82, 83]. Hydroxyl radical is the most reactive of ROS and can damage proteins, lipids, carbohydrates, and DNA. It can also start lipid peroxidation by taking an electron from polyunsaturated fatty acids.

Granulocytic enzymes further expand the reactivity of H_2O_2 via eosinophil peroxidase and MPO. In activated neutrophils, H_2O_2 is consumed by MPO. In the presence of chloride ion, H_2O_2 is converted to hypochlorous acid (HOCl). HOCl is

TABLE 2.1 Major endogenous oxidants.

Oxidant	Formula	Reaction equation
Superoxide anion	$O_2^{\bullet-}$	$NADPH + 2O_2 \leftrightarrow NADP^+ + 2O_2^{\bullet-} + H^+$ $2O_2^{\bullet-} + H^+ \rightarrow H_2O_2 + O_2$
Hydrogen peroxide	H_2O_2	$Hypoxanthine + H_2O + O_2 \rightleftarrows Xanthine + H_2O_2$ $Xanthine + H_2O + O_2 \rightleftarrows Uric\ acid + H_2O_2$
Hydroxyl radical	$^{\bullet}OH$	$Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH^- + {}^{\bullet}OH$
Hypochlorous acid	HOCl	$H_2O_2 + Cl^- \rightarrow HOCl + H_2O$
Peroxyl radicals	ROO•	$R\bullet + O_2 \rightarrow ROO\bullet$
Hydroperoxyl radical	HOO•	$O_2^- + H_2O \rightleftarrows HOO^{\bullet} + OH^-$

highly oxidative and plays an important role in the killing of pathogens in the airways [84]. However, HOCl can also react with DNA and induce DNA-protein interactions, produce pyrimidine oxidation products, and add chloride to DNA bases [85, 86]. Eosinophil peroxidase and MPO also contribute to the oxidative stress by modification of proteins by halogenations, nitration, and protein cross-links via tyrosyl radicals [87–89].

Other oxygen-derived free radicals are the peroxyl radicals ($ROO^{\bullet}$). The simplest form of these radicals is hydroperoxyl radical ($HOO^{\bullet}$) and it has a role in fatty acid peroxidation. Free radicals can trigger lipid peroxidation chain reactions by abstracting a hydrogen atom from a sidechain methylene carbon. The lipid radical then reacts with oxygen to produce peroxyl radical. Peroxyl radical initiates a chain reaction and transforms polyunsaturated fatty acids into lipid hydroperoxides. Lipid hydroperoxides are very unstable and easily decompose to secondary products, such as aldehydes (such as 4-hydroxy-2,3-nonenal) and malondialdehydes (MDAs). Isoprostanes are another group of lipid peroxidation product that are generated via the peroxidation of arachidonic acid and have also been found to be elevated in plasma and breath condensates of asthmatics [90, 91]. Peroxidation of lipids disturbs the integrity of cell membranes and leads to rearrangement of membrane structure.

Hydrogen peroxide, superoxide radical, oxidized glutathione (glutathione disulfide [GSSG]), MDAs, isoprostanes, carbonyls, and nitrotyrosine can be easily measured from plasma, blood, or bronchoalveolar lavage samples as biomarkers of oxidation by standardized assays.

2.3 Mitochondria as main source of ROS in autophagy signaling

Although the question is still far from being solved, there are at least other two issues that deserve to be considered. The first is why ROS are so rapidly produced. It would be more logical that a stimulus coming from the outside of the cell is transduced by a ROS-producing system located at, or nearby, the plasma membrane, such as the NOX complexes. Nevertheless, although attractive, this hypothesis has been verified only in macrophages upon bacterial infection, where ROS generated by NOX2 are indispensable for the recruitment of the microtubule-associated protein light chain-3 (LC3) on phagosomes that, thus modified, are degraded by autophagy to prevent pathogen escape [24]. Instead, a large amount of dataconverge to state that the mitochondria represent the principal source of ROS required for autophagy induction [92, 93], although they are not in close proximity to the plasma membrane. A possible explanation for this unexpected evidence is that nutrient deprivation suddenly results in energetic stress that, in turn, increases ATP demand and causes mitochondrial overburden to face up to adverse conditions. As a consequence, electron leakage and ROS production also increase. The second hypothesis is that a still uncharacterized factor could act as transducer, linking the upstream autophagic signal with mitochondria. A good candidate could be hexokinase II (HKII) that, by sequestering mammalian target of rapamycin complex-1 (mTORC1), could loosen its inhibition on the permeability transition pore and its ability to decrease ROS [94, 95]. In support of this hypothesis, it has been reported that the two protein kinases positively regulating HKII activity, Akt and myotonic dystrophy protein kinase, are negative modulators of autophagy [96, 97].

2.4 ROS and mitophagy

As principal sites of ROS production, mitochondria are the organelles that are able to turn on and tune autophagy. However, upon chronic impairment of mitochondrial function, ROS can be generated at high extent, thus shifting their role from bulk autophagy inducers into a self-removal signal for mitochondria through a selective process called mitophagy. This represents a fine mechanism of negative feedback regulation by which autophagy eliminates the source of oxidative stress and protects the cell from oxidative damage.

Although necessary, mitophagy represents an “extreme decision” for a cell subjected to nutrient deprivation because of at least two main reasons. The first reason is that mitochondria underpin ATP production that is fundamental upon carbon source limitation. The second reason lies in the fact that mitochondria are relatively large organelles that beforehand need to be fragmented to be properly recognized and engulfed within the autophagosomes [98]. Both these issues contribute to explain why mitochondria are in general refractory to undergo mitophagy, unless they are severely damaged. It has been proposed that under nutrient deprivation, mitochondria attempt to protect themselves from autophagic removal by promoting fusion and inhibiting fission events [99, 100]. The combination of these two inputs results in mitochondrial elongation that further impedes organelle engulfment within the autophagosomes and, concomitantly, maximizes ATP production [100]. Only upon prolonged starvation will mitochondria depolarize and become fragmented to assist their removal by mitophagy.

So far, at least two different molecular mechanisms underlying mitophagy have been described and characterized. The first one is mediated by NIX/Bnip3L (Bcl-2/adenovirus E1B 19-kDa-interacting protein 3, long form) [101, 102] an atypical BH3 protein that is able to directly recognize the autophagosome-sited GABA receptor-associated protein, a functional homolog of LC3 and, in turn, allow mitochondria to be removed. This is a "programmed" event, independent of any damage response that is required, for example, in mitochondrial elimination during erythrocyte differentiation [103, 104]. The second mechanism is activated for the selective dismissal of impaired or dysfunctional mitochondria. It is responsive to mitochondrial depolarization and is regulated by the PTEN-induced putative kinase-1 (PINK1) and Parkin [105], a ubiquitin E3 ligase whose mutations have been associated with a familial form of Parkinson's disease. PINK1 is a Ser/Thr kinase that translocates on the outer mitochondrial membrane where it is stabilized by low mitochondrial transmembrane potential, thereby acting as a real sensor of mitochondrial depolarization [106]. Here, PINK1 recruits Parkin [106] that ubiquitylates a number of outer mitochondrial membrane-located proteins, for example, the voltage-dependent anion channel-1 (VDAC1) [107]. Once ubiquitylated, these proteins are recognized by p62/sequestosome-1 (p62/SQSTM1, or simply p62) [107, 108], a ubiquitin-binding protein acting as a scaffold for several protein aggregates and triggering their degradation through the proteasome, or the lysosome pathway via autophagy [109, 110]. p62 contains an LC3 interacting region [111] that has been indicated as being fundamental for p62 to mediate mitophagy. Indeed, by means of this motif, p62 can bridge autophagy-targeted mitochondria to LC3 located on the autophagosome's surface, thereby driving their degradation. A role for Ambra1 in mitophagy, driven by its selective interaction with LC3 and independent from Parkin and p62, has been identified [112].

2.5 Production of ROS and their mechanisms of biological activities

ROS ($^1O_2$, $O_2^{\bullet -}$, H_2O_2, $^{\bullet}OH$, OH^-). However, other special ROS with biological effects exist. NO, for example, is a short-lived molecule with a free electron that regulates many physiological functions by itself, some of which are associated with development [113, 114]. Hydrogen peroxide is not as reactive as the hydroxyl radical yet the latter is readily generated when the former is in the presence of Fe^{3+} (Fenton reaction). Superoxide is also not very reactive but can react with NO to produce the very potent oxidant peroxynitrite [115]. Singlet oxygen, an electronically excited form of oxygen, is very reactive and produces clear effects on cells [116], but its biological relevance waits for more in vivo studies.

Phagocytic cells, such as macrophages and neutrophils, are also prominent sources of O_2^-. In the presence of invading pathogens like bacteria, phagocytic cells become activated and they generate O_2^-, which attacks the invading pathogens as a part of the inflammatory response [117]. Superoxide anion is also produced from xanthine by the enzyme xanthine oxidase along with uric acid as waste products of purine metabolism [118]. Other intracellular sources of the generation of ROS include reactions involving cytochrome P450 enzymes [119], peroxisomal oxidases [120], and NOXs [121].

Superoxide anion is important in the body because it generates other free radicals capable of causing cell injury [122, 123]. Superoxide anion is impermeable to the cell membrane and mainly affects enzyme function [124]. Its mechanism of toxicity involves disassembly of iron-sulfur (Fe-S) clusters in proteins through the inactivation of iron regulatory protein-1, causing release of iron and alterations of —SH residues [120]. Superoxide anion reacts with the potent vasodilator and cell-signaling molecule, NO, to form the toxic peroxynitrite ($ONOO^-$) product [123, 124]. Superoxide anion can also undergo dismutation to form hydrogen peroxide (H_2O_2), either spontaneously or when catalyzed by the enzyme SOD [122]. Although, H_2O_2 is not a free radical by definition, it is a biologically important oxidant because it selectively participates in hydroxyl radical generation, an extremely potent radical [124–126]. Hydrogen peroxide undergoes a reaction with metal ions like ferrous (Fe^{2+}) or cuprous (Cu^{2+}) to form ferric (Fe^{3+}) or cupric (Cu^{3+}) ions and hydroxyl ions, which is sometimes described as the Fenton reaction [63]. Also, because of its nonionized and low charged state, H_2O_2 has a long diffusion distance, since it readily diffuses through hydrophobic membranes as seen with the leakage of H_2O_2 from mitochondria [127]. H_2O_2 is broken down to O_2 and water by the antioxidant enzyme catalase [124, 127, 128]. In addition to catalase, GPx can also break down H_2O_2 and other peroxides that are formed on lipids within the body to yield water and GSSG [124, 127].

NO is synthesized through the enzymatic conversion of L-arginine to L-citrulline by nitric oxide synthase (NOS), which exists in three known forms of endothelial NOS (eNOS), inducible NOS (iNOS), and neuronal NOS, which can be upregulated under certain conditions to induce oxidative stress [129]. At physiological concentrations, NO controls mitochondrial respiration, causing reversible inhibition of respiration by altering cytochrome-*c* oxidase (Complex IV) activity, which is the terminal enzyme of the mitochondrial respiratory chain [130]. NO also binds to soluble guanylate cyclase in the vascular endothelium to control vascular tone [131]. iNOS is upregulated by oxidative stress, producing a burst of NO that far exceeds basal levels, which can cause significant cellular injury via different mechanisms: (1) NO may directly promote excessive peripheral vasodilation, resulting in vascular decomposition [131, 132], and (2) NO may upregulate the transcription of NF-κB, thus initiating an inflammatory signaling pathway that, in turn, triggers numerous inflammatory cytokines [132]. Peroxynitrite, obtained from the reaction of O_2^- and NO, has been reported to modify proteins with thiol groups resulting in the generation of nitrosothiols, which can disrupt metal-protein interactions and lead to the generation of other metal-derived free radicals [133]. Szabo et al.'s [133] study also showed that peroxynitrite can react with CO_2 to form nitrogen dioxide ($^{\bullet}NO_2$), a radical of less activity than peroxynitrite but of longer diffusion distance. The formation of superoxide anion is:

$$O_2 + e^- \xrightarrow{X} \text{Hypoxanthine} \xrightarrow{XO} \text{Uric acid} + O_2^{\bullet-} + H_2O_2$$

X : Xanthine; XO : Xanthine oxidase

Formation of peroxynitrite and nitrogen dioxide

$$NO + O_2^{\bullet-} \rightarrow ONOO^-$$

$$ONOO^- + CO_2 \rightarrow NO_2^{\bullet-} + CO_3^{\bullet-}$$

Formation of hydroxyl ions via Fenton reactions

$$Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH^{\bullet-} + OH^-$$

$$Fe^{2+} + H_2O_2 \rightarrow Fe^{2+} + OOH^{\bullet} + H^+$$

Hydroxyl radical ($^{\bullet}OH$) is formed not only by the interaction between hydrogen peroxide and the reduced forms of metal ions, i.e., Cu^{2+} and Fe^{2+}, but also by reduction of hydrogen peroxide as well as the interaction of superoxide with hydrogen peroxide [122, 124]. The hydroxyl radical is particularly unstable and is the most reactive of the free radical molecules. In addition, it is capable of reacting rapidly and nonspecifically with most biological molecules [124, 134]. Despite having very short half-lives of nanoseconds, they can cause severe damage to cell and other intracellular structures because they can cause covalent cross-linking of a variety of biological molecules. They cause cell damage by initiating chemical chain reactions such as lipid peroxidation, or by oxidizing DNA or proteins [126, 134–136]. Damage to DNA can cause mutations [136, 137] and possibly lead to cancer [137], if not reversed by DNA repair mechanisms [137]. Also, damage to proteins causes enzyme inhibition, denaturation, and protein degradation [134].

ROS are formed as unavoidable by-products of aerobic respiration and various other catabolic and anabolic processes [138] (Fig. 2.1).

For example, the respiratory chain produces essentially superoxide that can be converted to peroxide by SODs [139]. Among the enzymes that produce ROS by-products are (Fig. 2.1): fatty acyl-CoA oxidase, xanthine oxidase, cytochrome p450 systems, cyclooxygenases, and lipoxygenases [140, 141]. ROS are directly produced from oxygen by NOXs, a major family of enzymes whose catalytic subunits are encoded by Nox1–5 and Duox1–2 [69]. Although this activity was initially detected during phagocytosis, it is now known that these enzymes are broadly distributed among many tissues [69]. Interestingly, NOX enzymes are almost exclusive to multicellular organisms [142].

Due to the high reactivity of some ROS, the location where they are produced is critical for their biological effects (Fig. 2.1). Nonetheless, although it is common to think that ROS are cell autonomous signals produced within the affected cell, there are examples in which ROS holding low reactivity, such as hydrogen peroxide, appear to mediate intercellular communication. Paracrine communication could result when ROS are released from normal (i.e., myofibroblasts) or apoptotic cells and affect neighboring cells [143, 144]. It is frequently considered that hydrogen peroxide diffuses freely through biological membranes; however, its water-like electrostatic characteristics suggest a limited simple passive diffusion. It has been found that specific aquaporins, initially described as water channels that are present in all living cells, facilitate hydrogen peroxide diffusion across cell membranes [75] (Fig. 2.1). Diffusion of hydrogen peroxide also plays a role in the autocrine signaling mediated by NOXs (Fig. 2.1). The infusibility of ROS is a property that may contribute to determine the redox state of a community of cells or the propagation of ROS signals, mechanisms that could coordinate developmental events such as massive cell death.

2.6 Increased ROS production in photosynthesis during drought

The commonly considered sources of ROS shown in Fig. 2.2 potentially allow specificity because of their differences in chemical nature or location. Although drought may stimulate all three sources simultaneously, and ROS are frequently associated with damage, at least some of these pathways may act rather as damage limitation (protective) processes.

Much of the work on redox changes triggered by drought has focused on shoots. According to the dominant view of drought-induced oxidative stress in these organs, ROS production is increased by redox changes associated with photosynthesis. Even under optimal conditions, ROS can be produced at considerable rates inside the cell as part of metabolism

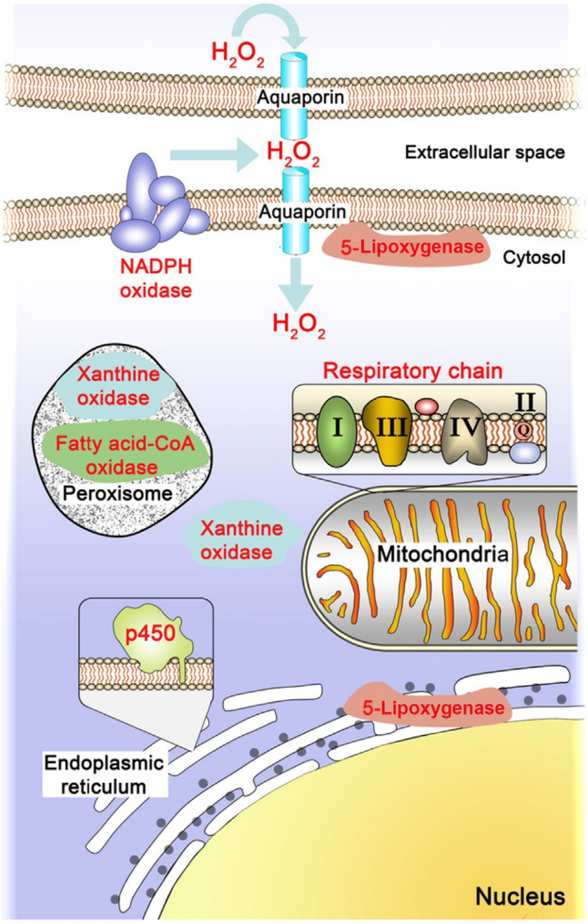


FIG. 2.1 Places in the cell where ROS are produced. Major organelles that are known to be sources of ROS are depicted. The activity of the respiratory chain in the mitochondria is responsible for most ROS produced in aerobiosis. On the other hand, the metabolic pathway that drives the degradation of long chain fatty acids (i.e., β-oxidation) in the peroxisome is also an important ROS source, though the amount produced depends on the activity of this metabolic pathway, a property that is cell-type specific. The function of ROS produced by cytochrome p450 or NADPH oxidases may be restricted to the area where they are located. Specific cytochrome p450 is involved in the synthesis and degradation of steroid hormones and retinoic acid, relevant molecules in development. Peroxisomal or cytosolic xanthine oxidase is an enzyme that produces ROS from molecular oxygen, whose best-characterized function is in the final catabolism of purines. 5-Lipoxygenase, an enzyme involved in the synthesis of leukotrienes, can be found associated with membranes or with the nuclear envelope. ROS can also function as paracrine signals, as hydrogen peroxide can cross from one cell to another through aquaporins.

[145]. In terms of production of H_2O_2, the most stable of the major ROS species, the chloroplast, continues to receive the most attention; however, in many conditions in C_3 plants, the rate of H_2O_2 production may actually be higher in the peroxisomes [146]. Peroxisomal H_2O_2 production in the green tissues of C_3 plants is largely a result of the activity of glycolate oxidase. This enzyme is an essential part of the photorespiratory recycling pathway that is initiated by oxygenation of ribulose-1,5-bisphosphate (RuBP) in the chloroplast [147]. Glycolate production is considered to be accelerated during drought as intercellular CO_2 drops as a result of drought-induced stomatal closure. This favors RuBP oxygenation [148] and hence increased peroxisomal H_2O_2 production (Fig. 2.2B, site 1).

In the chloroplast, restrictions over reductant and ATP consumption during drought may also favor ROS production at two distinct sites within the electron transport chain. First, decreased availability of other oxidants for the chain may promote electron flow to O_2 in the Mehler reaction, thus stimulating superoxide and H_2O_2 production and accelerating the water-water cycle ([149]; Fig. 2.2B, site 2). Second, any overreduction of the electron transport chain is expected to enhance the probability of singlet oxygen generation in photosystem II ([150]; Fig. 2.2B, site 3).

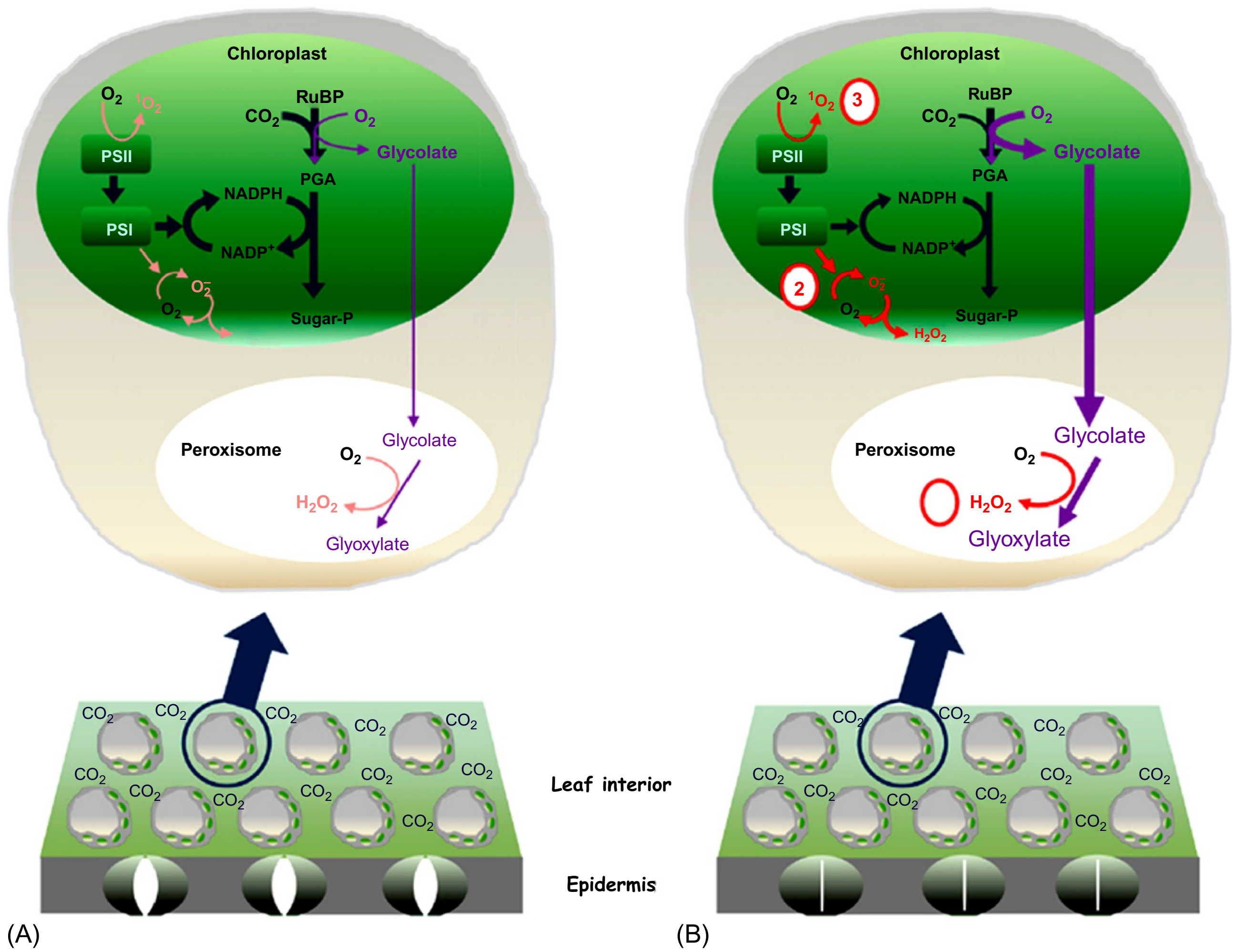


FIG. 2.2 Current concepts of how drought increases the generation of ROS in photosynthesis. (A) Cartoon of leaf section in well-watered plants in which relatively high intercellular CO_2 concentrations (Ci) allow efficient regeneration of terminal oxidants and limit RuBP oxygenation. (B) Drought-induced stomatal closure restricts CO_2 uptake, favoring photorespiratory production of H_2O_2 in the peroxisome (1) and possibly favoring production of superoxide and H_2O_2 (2) or singlet oxygen (3) by the photosynthetic electron transport chain. *PGA*, 3-phosphoglyceric acid.

According to this view, production of superoxide and H_2O_2 at both sites 1 and 2 is associated with limitation of the accumulation of reduced intermediates and hence a decrease in the probability of singlet oxygen generation at site 3 (Fig. 2.2).

Several genes encoding expansins are among genes that are upregulated at an early stage or in response to moderate drought [151]. Some of these adjustments in cell wall structure may involve ROS and thus one or more of several types of enzymes that are localized at the cell surface or apoplast and that use different reductants and cofactors to produce either superoxide or H_2O_2 (Fig. 2.3).

These include NOXs, amine oxidases, polyamine oxidases, oxalate oxidases, and a large family of class III heme peroxidases [152–155]. Class III heme peroxidases may either use H_2O_2 to oxidize apoplastic substrates or reductants to produce superoxide from O_2 [152]. Although most of these ROS-producing enzymes have been implicated in pathogenesis responses [152, 154, 156, 157], the roles of many of them in drought are less clear. However, enzymes such as oxalate oxidases may play important roles in the acclimation of root growth to drought [158]. In addition, the plasma membrane is rich in stress perception proteins such as receptor-like kinases that fulfill important roles in drought tolerance and cell wall function [159, 160]. Many of these are regulated at the level of expression by changes in the cell redox state [161].

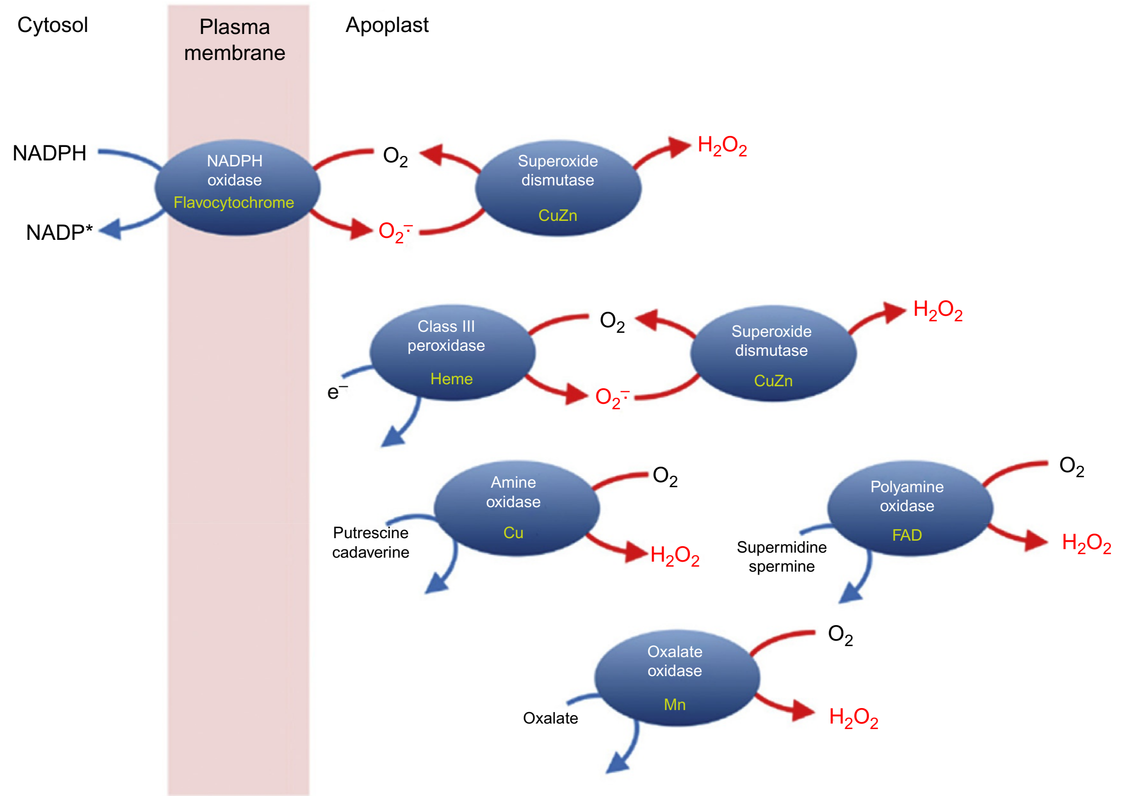


FIG. 2.3 Multiple ROS-producing enzymes at the cell surface/exterior. Enzymes are shown in *blue* and their redox cofactors are indicated in *yellow*. Class III peroxidases may accept electrons from several types of compounds to generate superoxide, but in many cases their physiological reductant is not established [152].

2.7 ROS elimination

Levels of ROS are not only determined by production, but also by the rate of ROS degradation or inactivation. In general terms, the ultimate effect of antioxidants is to decrease the amount of active ROS. Cells have many ways to respond against ROS, including enzymatic and nonenzymatic antioxidants. It is the balance between the production and degradation of ROS that maintains the cellular homeostasis. Common nonenzymatic antioxidants are GSH and Trx [162, 163]. GSH synthesis is catalyzed by the sequential action of γ-glutamyl cysteine synthetase and GSH synthetase [164]. Once GSH is oxidized (GSSG), the reduced form can be regenerated by GSH reductase. The balance between GSH and GSSG is a way to determine the redox state within the cell. On the other hand, Trx, as well as Grx, are small proteins containing an active site with a redox-active disulfide [162]. These proteins maintain a reduced intracellular redox state in mammalian cells by the reduction of protein thiols. Two Trx and three Trx reductases (TrxR) are present in mammals, each with a distinct intracellular location [165]. Trx1 and TrxR1 are cytosolic or nuclear proteins, whereas Trx2 and TrxR2 are targeted to the mitochondria. Of the two Grx present in mammals, Grx2 appears to be in mitochondria and the nucleus [166].

2.8 Types of reactive oxygen species

ROS are produced from molecular oxygen as a result of normal cellular metabolism. ROS can be divided into two groups: free radicals and nonradicals. Molecules containing one or more unpaired electrons and thus giving reactivity to the molecule are called free radicals. When two free radicals share their unpaired electrons, nonradical forms are created. ROS in high concentrations have damaging actions, but in lower concentrations they can act as signaling molecules (Fig. 2.4).

ROS generated by the activation of enzymes such as NOX, xanthine oxidases, uncoupled NO synthases, and other sources such as arachidonic acid metabolizing enzymes, lipoxygenases and cyclooxygenases, cytochrome P450s, peroxidases, and other hemoproteins, as well as ROS generated by mitochondria, seem to play various roles in the cellular signaling network under different physiological and pathophysiological conditions. Various cellular antioxidant systems oppose ROS load thereby limiting not only cellular damage, but also contributing to ROS-dependent signaling. The delicate

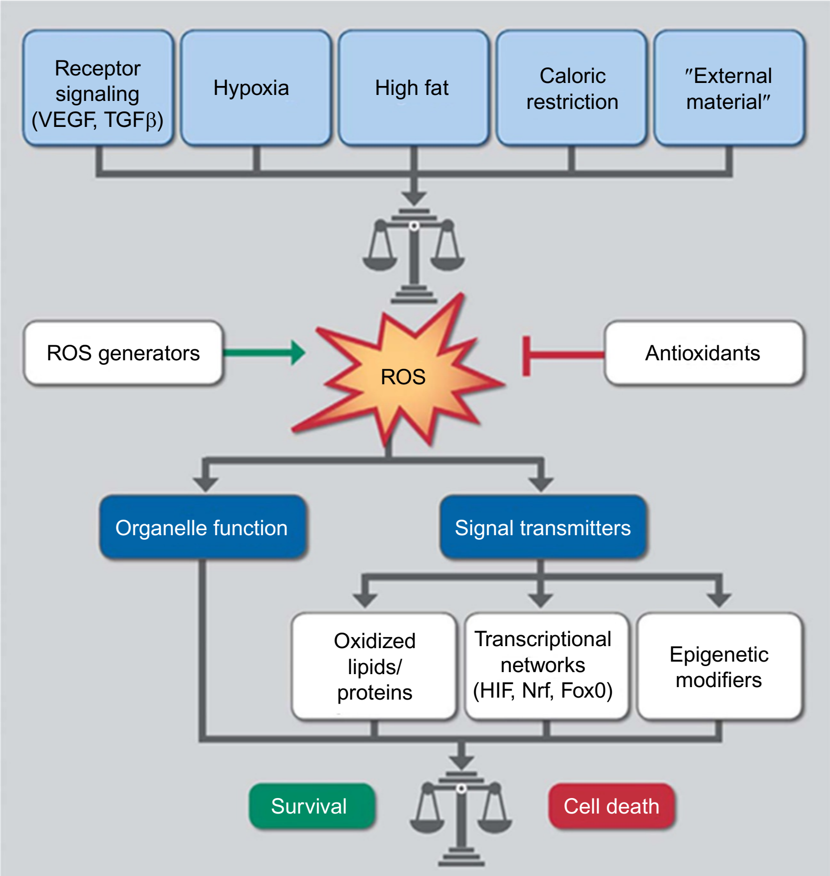


FIG. 2.4 Summary scheme of ROS acting as signaling molecules in different disease settings but also in physiological processes.

balance between ROS generation and ROS scavenging is disturbed by the different types of stress factors like salinity, drought, extreme temperatures, heavy metals, pollution, high irradiance, pathogen infection, etc. (Fig. 2.5A).

Most ROS are generated as by-products during mitochondrial electron transport. In addition, ROS are formed as necessary intermediates of metal-catalyzed oxidation reactions. Atomic oxygen has two unpaired electrons in separate orbits in its outer electron shell. This electron structure makes oxygen susceptible to radical formation. The sequential reduction of oxygen through the addition of electrons leads to the formation of a number of ROS, including: superoxide, hydrogen peroxide, hydroxyl radical, hydroxyl ion, and NO (Fig. 2.5B). The major ROS include: $^1O_2$, superoxide ($O_2^{\bullet -}$), hydrogen peroxide (H_2O_2), hydroxyl anions (OH^-), hydroxyl radicals ($OH^\bullet$), and hypochlorous acid (HOCl). Superoxide is produced by NOX/xanthine oxidase-derived reduction of molecular oxygen, uncoupled eNOS, or mitochondrial ETC. Superoxide is rapidly dismutated to H_2O_2 by SOD. However, in the presence of NO, $O_2^{\bullet -}$ rapidly reacts with NO, resulting in the formation of highly reactive peroxynitrite ($ONOO^-$), which is three to four times faster than dismutation of $O_2^{\bullet -}$ to H_2O_2. H_2O_2 can change to highly reactive HOCl at the inflammatory sites by an enzyme known as MPO, which is abundantly expressed in neutrophils. H_2O_2 can also change to the highly toxic $OH^\bullet$ in the presence of Fe^{2+} by Fenton's reaction. H_2O_2 is scavenged to H_2O and O_2 by catalase, GPx, or Prx antioxidant enzymes. Prx uses thioredoxin (Trx) to detoxify H_2O_2. These are lethal and cause extensive damage to biomolecules such as lipids, proteins, and DNA (Fig. 2.6) and thereby affect normal cellular functioning [167].

Bivalves are able to survive in a wide range of oxygen concentrations ranging from anoxic to high levels of dissolved oxygen. Variations in this ability have been proposed as an index of environmental stress. Oxyradicals ($O_2^{\bullet -}$, H_2O_2, $^\bullet OH$) can be highly toxic to aquatic organisms often resulting in lipid peroxidation in membranes, altered pyridine nucleotide redox status, and DNA damage. Moreover, many xenobiotics are capable of modulating oxidative stress either by acting directly as redox cycling compounds (e.g., menadione) or as a consequence of biotransformation to quinones, which are redox cycling (e.g., benzo(*a*)pyrene). The microsomes appear to be an important subcellular location for such activity since this is the site of cytochrome P450 induction and catalysis. An attractive in vitro system for studying the production of oxyradicals in *Mytilus edulis* lysosomes using the laser dye dihydrorhodamine 123 is available.

FIG. 2.5 (A) Various causes responsible for the generation of ROS. (B) Electron structures of common ROS. Each structure is provided with its name and chemical formula. The ● designates an unpaired electron.

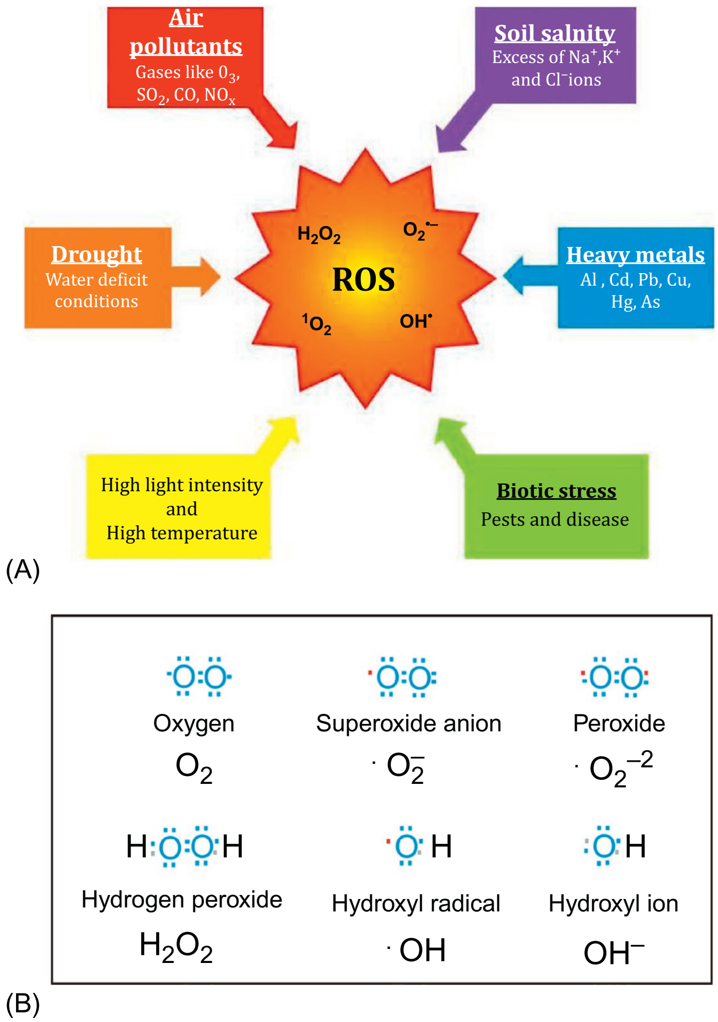
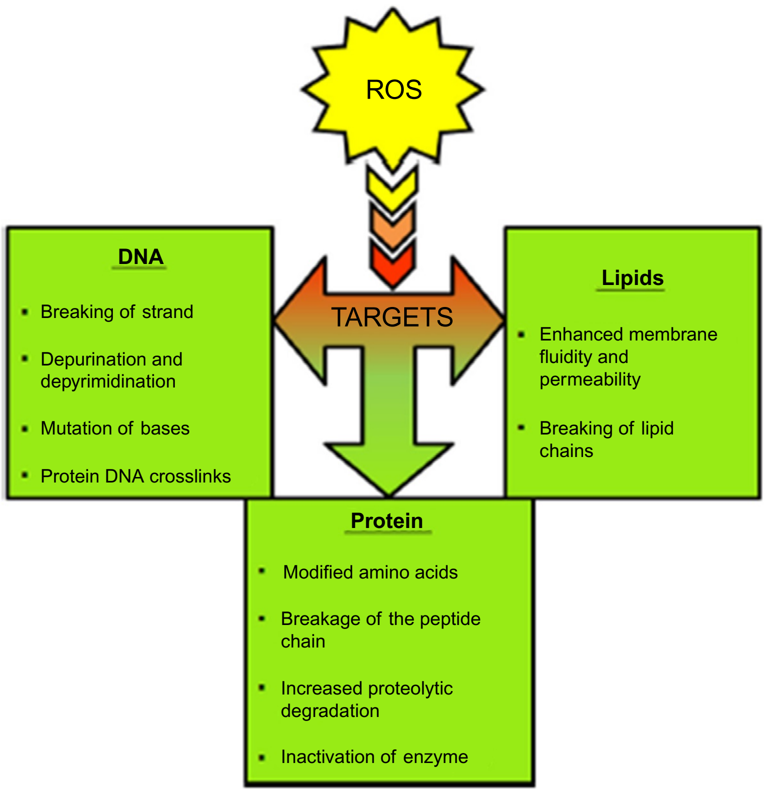


FIG. 2.6 Reactive oxygen species (ROS)-induced oxidative damage effects to tissues, lipids, proteins, and DNA.



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